

MEHUL KATARIA

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SUMMARY

CS (AI & ML) student with internship experience in Python Development and Data Analytics. Proficient in C++, Python, and Machine Learning, with a strong problem-solving background (150+ solutions across platforms). Seeking AI / ML and Software Development internships.

EDUCATION

Jodhpur Institute of Engineering and Technology

Jodhpur, India

B.Tech in Computer Science (AI & ML)

2023 – 2027 *Current CGPA: 9.48/10*

EXPERIENCE

Local Business

Freelance

Data Analyst Job

May 2024 – October 2024 *Data Visualization: Empowering Business*

- Designed interactive dashboards using **Matplotlib** and **Seaborn** to visualize key business KPIs for senior leadership.
- Performed data cleaning and preprocessing on raw datasets using **Pandas**, ensuring 99% data accuracy.
- Transformed raw operational data into actionable business insights, driving a 15% increase in operational efficiency.

Devnex Technologies, Bangaluru, Karnataka

Remote

Python Development Intern

May 2025 – June 2025 *Machine Learning Projects*

- Developed and maintained scalable RESTful APIs using **Django** and **Flask** to support web application features.
- Automated data processing pipelines using **Python scripts**, reducing manual entry time by 40%.
- Collaborated with the engineering team to debug and optimize existing codebases, improving application performance.

PROJECTS

Image Super-Resolution using GANs

Deep Learning & Computer Vision

Deep Learning Engineer

2025 *A generative AI model capable of upscaling low-resolution images by 4x with high perceptual quality.*

- Implemented **SRGAN** using **PyTorch** to recover fine texture details often lost in standard MSE-based methods.
- Engineered a custom **Perceptual Loss function** (combining Content & Adversarial loss) to generate photo-realistic natural images.
- Trained on the **DIV2K dataset**, achieving a **PSNR of 29.22** and **SSIM of 0.846** on the Set5 validation benchmark.
- Tech Stack:** Python, PyTorch, OpenCV, Torchvision, NumPy.

Intelligent Document Summarizer & Q&A System

NLP & GenAI

NLP Engineer

2024 – Present *An advanced NLP platform for summarizing PDFs and answering context-aware user queries.*

- Implemented abstractive summarization using **BART-Large-CNN** with sliding window chunking for long documents.
- Built a **RAG pipeline** using **Sentence Transformers** and **RoBERTa** for semantic search and precise Q&A.
- Developed a real-time **Streamlit** interface with **PyMuPDF** text extraction and answer confidence scoring.
- Tech Stack:** Python, Streamlit, Hugging Face, PyTorch, RAG Pipeline.

Service Desk Ticket Classifier

Deep Learning & Automation

Deep Learning Engineer

2025 *An intelligent system that streamlines support operations by classifying and routing tickets automatically.*

- Developed a **Convolutional Neural Network (CNN)** to classify support tickets into 5 distinct categories (Technical, Billing, etc.).
- Implemented an **Automated Email Routing** system using SMTP to instantly direct tickets to the relevant department based on AI predictions.
- Built an interactive user interface using **Streamlit**, enabling real-time ticket entry, classification, and status tracking.
- Tech Stack:** Python, PyTorch, Streamlit, SMTP, NLTK, Pandas.

CERTIFICATIONS

Data Analytics & Visualization – Accenture, North America (Forage)

Database Design and ER modeling & C programming – InfoSys Springboard

Software Engineering Job Simulation – Electronic Arts (Forage)

CS205: Building with Artificial Intelligence & CS201: Data Structures – Saylor Academy

NPTEL Elite Silver: Ethics in Engineering, Soft Skills Development (Top 2%)

AI Apprentice – Intel & Digital India Foundation

Introduction to Cybersecurity – Cisco Networking Academy

ACHIEVEMENTS

GATE DA 2026: Qualified the Graduate Aptitude Test in Engineering in Data Science & Artificial Intelligence 2026.

Coding Profile: Achieved **5-Star Badge** in Python & C on HackerRank; Solved **150+ DSA problems** on LeetCode/HackerRank.

CodeVita: Cracked **TCS CodeVita Season 13** round 1 against **500k+** participants.

Hackathon Winner: Secured **2 wins** and achieved Finalist status in **10+ national-level hackathons**, competing against 500+ teams.

Aptitude: Secured **2nd Rank** in Inter-college Quantitative Aptitude competition.

Academic Excellence: Maintained a cumulative **CGPA of 9.48** in B.Tech Computer Science (AIML).

SKILLS

Languages: Python, C, C++, SQL, Javascript, TypeScript

Frameworks: React, Tailwind CSS, Django, Flask, Streamlit, OpenCV, Pandas, NumPy, TensorFlow, Keras, Scikit-Learn, NLTK

Web Technologies: HTML, CSS

Databases: MySQL, MongoDB

Tools & Platforms: Git, GitHub, VS Code, RESTful APIs